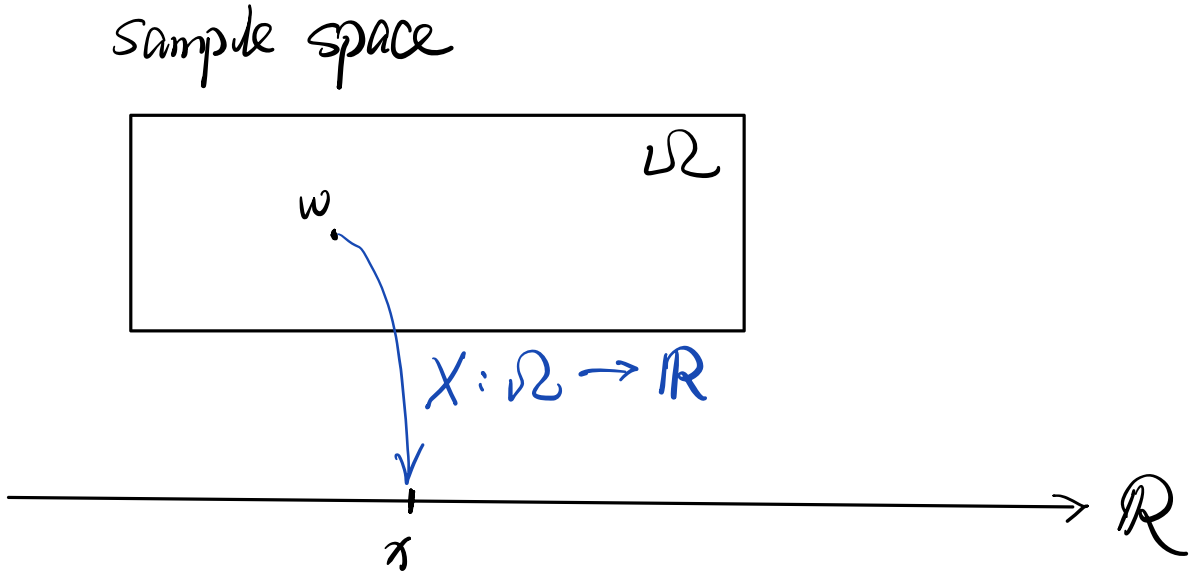
Last Time



X is a discrete random variable if the range of X is countable.

PMF : $p_X(x) = P(X = x)$

CDF : $F_X(x) = P(X \leq x)$

Recall the following example from *Lecture 07*.

Example. Suppose a fair coin is tossed three times. The sample space is

$$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}.$$

Define the random variable

$$X : \Omega \to \mathbb{R}$$

by letting $X(\omega)$ be the number of heads in the outcome ω. Thus,

$$\begin{aligned} X(hhh) &= 3, \quad X(hht) = 2, \quad X(hth) = 2, \quad X(thh) = 2, \\ X(htt) &= 1, \quad X(tht) = 1, \quad X(tth) = 1, \quad X(ttt) = 0. \end{aligned}$$

The range of X is

$$\{X(\omega) \mid \omega \in \Omega\} = \{0, 1, 2, 3\}.$$

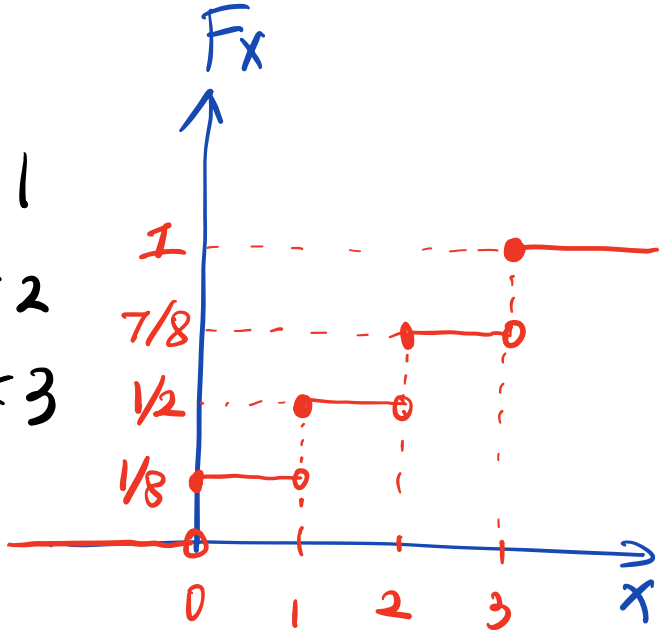
Since the range is finite, X is a **discrete random variable**.

We derived the probability mass function (PMF) of X:

$$p_X(x) = \begin{cases} \frac{1}{8}, & x = 0, \\ \frac{3}{8}, & x = 1, \\ \frac{3}{8}, & x = 2, \\ \frac{1}{8}, & x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Find and sketch the CDF of random variable X.

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1/8, & 0 \le x < 1 \\ 1/8 + 3/8, & 1 \le x < 2 \\ 1/8 + 3/8 + 3/8, & 2 \le x < 3 \\ 1, & 3 \le x \end{cases}$$



Recall the definition of CDF.

Cumulative Distribution Function

The **cumulative distribution function** (CDF) of random variable X is

$$F_X(x) = P(X \leq x)$$

where $x \in \mathbb{R}$ and $P(X \leq x)$ is the probability of the event $\{X \leq x\} = \{\omega \in \Omega \mid X(\omega) \leq x\}$.

Remarks

- For a discrete random variable, probability is concentrated at individual points. The CDF accumulates these probability masses, producing a step function.
- It should be noted that CDF is defined for any random variables, not necessarily discrete random variables.

Does every random quantity take only a countable number of possible values?

Example. Suppose a number is selected *uniformly at random* from the interval $[0, 1]$. Let

$$X = \text{the selected number.}$$

The possible values of X are $R_X = [0, 1]$. Unlike the range of a discrete random variable, $[0, 1]$ is uncountable. Hence, X is not a discrete random variable.

- What is the probability that the selected number is exactly 0.3, that is, $P(X = 0.3)$? $= 0$
- What is $P(X = 0)$? $= 0$
- What is $P(X = x)$ for any $x \in [0, 1]$? $= 0$
- What is $P(X = 2)$? $= 0$ (impossible event $\{X = 2\}$).

Event A is impossible $\longrightarrow$ $P(A) = 0$

Random quantities such as **waiting time**, **temperature**, **voltage**, and **measurement error** are naturally modeled as taking values continuously over an interval. This motivates another important class of random variables: **continuous random variables**.

CONTINUOUS RANDOM VARIABLES

Discrete Random Variable	Continuous Random Variable
Probability mass at individual points	Probability described over intervals
$\Downarrow$	$\Downarrow$
CDF has jumps	CDF is continuous

Remark

For the continuous random variables considered in this course, probability is described over intervals rather than concentrated at individual points. Their CDFs are continuous functions.

Recall:

- Cumulative distribution function (CDF):

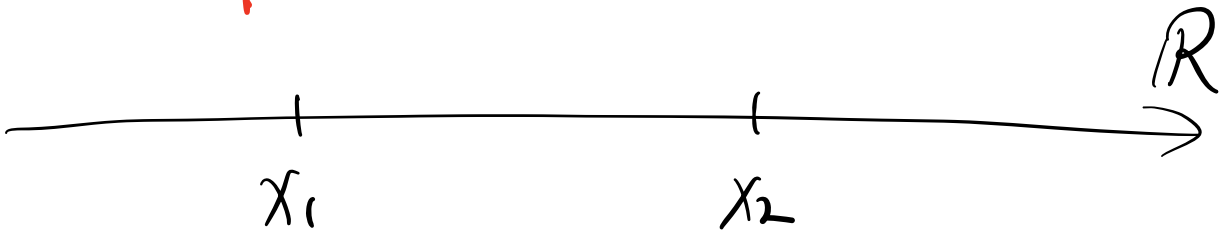
$$F_X(x) = P(X \leq x).$$

- A continuous function has no jumps, breaks, or holes in its graph.

Some Properties

For ***any*** random variable X,

- $F_X(-\infty) = 0$
- $F_X(\infty) = 1$
- $P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$



Example The cumulative distribution function of the random variable X is

Recall: $F_X(x) = P(X \le x)$

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{4}, & 0 \le x \le 4, \\ 1, & x > 4. \end{cases}$$

Sketch the CDF of X and calculate the following probabilities:

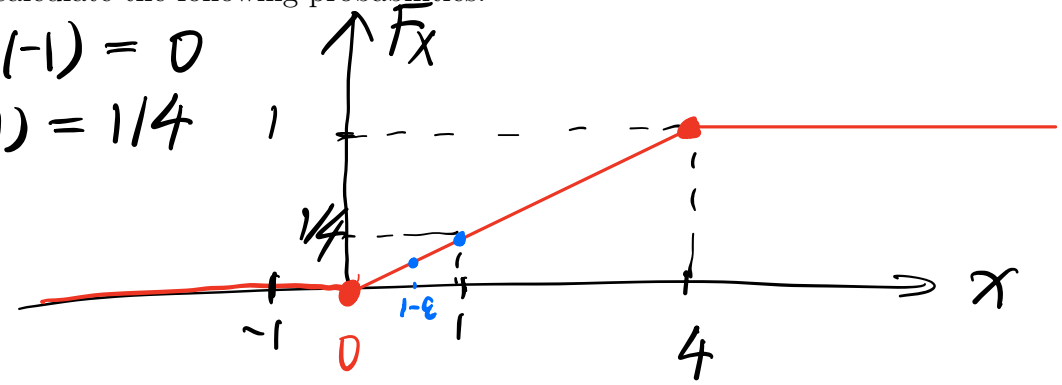
(a) $P(X \le -1) = F_X(-1) = 0$

(b) $P(X \le 1) = F_X(1) = 1/4$

(c) $P(2 < X \le 3)$

(d) $P(X > 1)$

(e) $P(X = 1)$



(c) $P(2 < X \le 3) = F_X(3) - F_X(2)$

$= 3/4 - 2/4 = 1/4$

(d) $P(X > 1)$: probability of the event $\{X > 1\}$

Since $\{X > 1\}^c = \{X \le 1\}$

Complement

$\therefore P(X > 1) = 1 - P(X \le 1)$

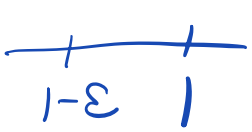
$= 1 - 1/4 = 3/4$

(e) $P(X = 1)$: probability of the event $\{X = 1\}$

For some small $\varepsilon > 0$, we have

$\{X = 1\} \subset \{1 - \varepsilon < X \le 1\}$

$\therefore P(X=1) \leq P(1-\varepsilon < X \leq 1)$



$= F_X(1) - F_X(1-\varepsilon)$

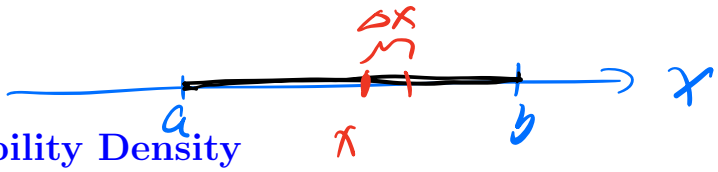
$= \frac{1}{4} - \frac{1-\varepsilon}{4}$

$= \frac{\varepsilon}{4} \quad (\forall \varepsilon > 0)$

$P(X=1) \leq \frac{\varepsilon}{4} \quad (\forall \varepsilon > 0)$

$\therefore P(X=1) \leq \lim_{\varepsilon \to 0} \frac{\varepsilon}{4} = 0$

$\therefore P(X=1) = 0.$



Motivation: Mass Density and Probability Density

Consider a thin bar extending from $x = a$ to $x = b$. Let

$$\rho(x) = \text{linear mass density at position } x,$$

where $\rho(x)$ may have units such as kg/m.

The value $\rho(x)$ is *not* the mass at the single point x. Instead, the mass contained in an interval $[c, d] \subset [a, b]$ is

$$M_{[c,d]} = \int_c^d \rho(x)\, dx.$$

The total mass of the bar is

$$M = \int_a^b \rho(x)\, dx.$$

For a small interval of length Δx,

$$\Delta M \approx \rho(x)\Delta x.$$

This motivates a similar idea for continuous random variables.

Probability Density

For a continuous random variable X, let

$$f_X(x) = \text{probability density at } x.$$

Just as $\rho(x)$ is not the mass at a single point, $f_X(x)$ is *not* the probability at a single point. Instead, probability over an interval is obtained by integrating the probability density:

$$P(c < X \le d) = \int_c^d f_X(x)\, dx.$$

For a small interval of length Δx,

$$P(x < X \le x + \Delta x) \approx f_X(x)\Delta x.$$

The analogy can be summarized as follows:

Mass along a bar	Probability distribution
$\rho(x)$	$f_X(x)$
mass density	probability density
$\int_c^d \rho(x)\, dx$	$\int_c^d f_X(x)\, dx$
mass in $[c, d]$	$P(c < X \le d)$
$\int_a^b \rho(x)\, dx = M$	$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$

mass density $\longrightarrow$ integrate to obtain mass

probability density $\longrightarrow$ integrate to obtain probability.

Continuous Random Variable & Probability Density Function

A random variable X is called a ***continuous random variable*** if there exists a nonnegative function

$$f_X : \mathbb{R} \to [0, \infty)$$

such that, for any $a < b$,

$$P(a < X \leq b) = \int_a^b f_X(x)\, dx.$$

The function f_X is called the ***probability density function (PDF)*** of X.

The probability density function f_X satisfies the following properties:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.

2. The total area under the PDF is one:

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1.$$

3. The CDF of X is given by

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

4. The probability at any individual point is zero:

$$P(X = x) = 0, \qquad x \in \mathbb{R}.$$

   Consequently, the inclusion or exclusion of endpoints does not affect the probability of an interval:

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b).$$

5. At points where the CDF F_X is differentiable, the PDF can be obtained from

$$f_X(x) = \frac{d}{dx} F_X(x).$$